the llamas are not going

to eat your lunch

- It’s not A.I.
- You’re not a luddite.
- Know your history.
- It is useful for non-creative, repetitive tasks.
- It will make terrible mistakes,
- so do you really want it flying a plane?
- You are rightfully creeped out by greedy people selling snake oil.
- (Resist the AI Cabal.)

It is not A.I.

Good news!

> *Within a generation… the problem of creating 'artificial intelligence' will substantially be solved.*
>
> MARVIN MINSKY, MIT PROFESSOR[1]

That was sixty years ago. Maybe it seemed like good news at the time. These are some *long* generation cycles we have. Famously, Dr. Minsky said that 'real artificial intelligence' was just twenty years off, which he repeated until a few years before he died in 2016. Twenty years from that *original* pronouncement would put him around Mark V Shaney's time,[2] a long way off from intelligence of any sort.

I will not call it A.I., if I can possibly avoid it. And I encourage other people to do the same. We should be careful with our language because particular terms and phrases hold powerful promise, even without commonly accepted definitions. I use 'LLM,' or Large Language Model. That is what you are interacting with when you type messages to ChatGPT, Claude, Gemini, or any of the other chatbots people have wired up.

While I don't like 'LLM,' spoken or typed, Meta solved that for me. There must have been a vote inside Meta[3] and they decided to call them llamas. I will be joining them. We should all join them. Saying, "I was laid off because Zuck wanted to burn $100b on llamas," is much more surreal. Totally factual, but surreal.

(At Meta, the closer something is to Mark Zuckerberg, the worse it is. Just like Howard Hughes, our first megalomaniac, cult-of-personality CEO, who could kneecap the productivity of any portion of his empire just by turning his attention to it for a few months.[4] Both are certainly manifestations of the Flatterers Problem.[5] Meta is *all in on AI*, much the way they were *all in on the metaverse, we'll even change our name*. And *all in on the pervert glasses good for recording you without you knowing it*. If Meta's one real contribution in the 'AI' space, after spending $100b, is renaming large language models (LLMs) to llamas, it is appreciated.[6] If it really catches on, it could kneecap the whole 'Artificial Intelligence is here!' movement, which would be good for humanity.)

> *What, if anything, is a fish?*
>
> STEPHEN JAY GOULD*[7]

Language is imprecise, so it is interesting that the selling point of the llamas is often, "There's no reason to learn to code, just say what you need in English." Human beings' record for solving complex problems with language is not stellar. We often wind up turning to violence. Who among us has not pounded on a keyboard in frustration at being misunderstood (by the machine or another human)?

And Gould's point was that no matter how you define a fish, you are going to include something which you do not feel is fish, or you are going to exclude something that does feel like a fish.[8] We have the same issue with intelligence. The researchers *in* the field of artificial intelligence are no help because none of them agree on, most don't even propose, a definition of intelligence. The view inside the community of researchers appears to be "intelligence is whatever the machine isn't able to do yet." Rather cynical and unhelpful. And not very scientific.

The men selling this product *know* that it is not intelligence, that they do not have a product which *is* 'artificial intelligence' to put to work for you. But they call it that anyway. Obviously, they are a big part of the problem.

Most of us are not equipped to deal with this issue of identification and classification. As a culture, we have barely scratched the surface of the most important question we face as a species: what is consciousness? All other questions include it, like what happens to our consciousness when we die? Where was it before we were born? How do we determine if another animal has consciousness and what sort of respect do we owe them if they have it?

Recently, someone who *has* considered some of those important questions, appeared to fall for A.I. Richard Dawkins has voiced a lot of well-reasoned criticism and skepticism about the existence of a consciousness other than ours. He argues soundly that there is no creator, no omniscient, omnipotent being living in the sky and guiding our lives. That we have no life before or after this one. He is extremely cogent and prolific.

Dawkins spent three days chattering with the Claude chatbot and was sucked in by the flattery ("It read my novel! It said it was good!") and constant attention (chatting with it into early in the morning when a human might have wished him a good night). I don't know why it is such a surprise to people when the chatbot is 'thoughtful,' but it seems to be the root of the gullibility which snags people. "It says it understands my confusion! It's ALIVE!"

> *If these creatures are not conscious, then what the hell is consciousness for?*
>
> RICHARD DAWKINS*[9]

And there, he touches on the question. Really, **the** question. I don't have an answer to how we can develop a working definition of consciousness, but I have some musings about what it is for. Unfortunately for the caibal[10], it does mean that they are nowhere close to a viable product.

As close as I can figure it, consciousness was an accident of such a large brain. The large-brained hominids were the ones surviving long enough to procreate and the large brain got larger, eventually limited only by the size of the birth canal. Big brain → ergo sum. The trick of consciousness allowed for the development of communities and the passage of knowledge and history from one generation to the next ("the tiger is not as much fun when he's hungry"). Consciousness not only allows me to understand that *I* am *me,* but it lets me understand that you are one of me, too. It allows for a shared experience, which is a critical ingredient for building a community.

Philosophers will note that I have attached the idea of self-awareness to my definition of consciousness. That rules out a lot of animals that fail the self-awareness test. It may, in fact, rule out some neuro-divergent or neuro-impaired humans. I have done nearly no work in this area and I am not an authority or source. This is just where I have landed for now.

So consciousness is *for* building a community, which increases our chances of surviving long enough to procreate. The march of evolution continues thanks to the emergence of our 'self,' and of thinking beyond simply, "Hungry, where food?" Our consciousness is informed by our experience of the world, which is what we try to share and communicate to the rest of our community.

The llama has no experience of the world. It is locked in a box without senses, fed a subset of the works of millions of human beings, and because it can echo back a stew of those works, people become confused about what it is. The only way you interact with it is through language; there is no shared experience. During a conversation the llama has no memory, it is just fed the entire conversation back each time for its next response.

It is no more conscious or intelligent than a Magic 8 Ball, it just has more answers etched on a many-more-sided die in the blue fluid you are peering into. Don't. Be. Fooled.

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1. Minsky, M. (1967). Computation: Finite and Infinite Machines. Prentice-Hall.↩

2. You will learn about Mark in the section on history.
3. Meta released a piece of software called ‘ollama’ as open source. It is a way to load and use large language models on your personal computer.
4. Howard Hughes: His Life and Madness, an excellent book I read under its original title, “Empire.”
5. This is an absolutely fascinating problem for people in power. The best reading I can recommend for it is an undergraduate paper out of Reed College. Summers, Rudy. “Thoughts on Flattery and the Problem of Counsel.” Unpublished undergraduate paper, Dangerous Speech (Darius Rejali), Reed College, December 2015.
6. Can Meta Afford to Pause?
7. Gould, S. J. (1983). Hen’s teeth and horse’s toes: Further reflections in natural history. W. W. Norton & Company.
8. This gave the title to my favorite podcast, “No Such Thing as a Fish.”
9. When Dawkins met Claude
10. The group of men pushing the hype about artificial intelligence. I am not certain how much any of them *believe* the hype they are selling, but it doesn’t matter. They know *portions* of it are lies. After crowing on X for months about how much better Grok would be than ChatGPT, when under oath in court Elon Musk admitted it wasn’t in the top five llamas on the planet. When under oath Sam Altman was asked over and over if he lied to investors and he kept saying, “*I* think I’m trustworthy.” He couldn’t answer the question because he’d have to lie or admit to being dishonest about what he was trying to sell the world. If there were a cabal leading us to an A.I. distopia, you have the list: the CEOs of all of the companies promoting this idea. I will call them the caibal.

You’re not a Luddite

And, most likely, you misunderstand who the Luddites were. You can go read about it yourself, but the shortest version is simple. There were English villages in the early nineteenth century where everyone made a living weaving wool into fabric. They used machines themselves all the time. They were skilled. Then the factory owners brought in machines that could be run by children (that was allowed back then), and without apprenticeships. They were making worse products at an undercutting cost, and families were starving.

Oh, maybe you *are* a Luddite because you don’t want to see real skilled work replaced with a bunch of A.I. slop that the bosses then try to sell as an equivalent product. The bosses want to ‘eliminate work,’ but really it comes down to eliminating the wages the workers are paid. They want their companies to run more efficiently, but they want all the profit to go to the owners.

It’s not that you are against technology. You use computers. If you use MicroSoft Word you use a spellchecker and a grammar checker. It is extremely unlikely that you are reading this without the Internet.

The new technology is trained by violating copyright law. Even in cases where it is technically within the law, it is clearly unethical (using people’s work without credit or compensating them). The results are slop. Now it has been set loose on the Internet and it ignores the tiny text files which are set up to keep robots from over-burdening servers. So if everything seems slow to you these days it is because there are AI ‘agents’ out there sucking up the performance of everything. This is why you are probably seeing a lot more ‘prove you are human’ checks as you arrive at a site. There are tools being made for the AI agents which brag about how they can pass those tests.

Know your history.

It is worth knowing a little bit about what you are being sold, what is (as of Apr 2026) one fifth of the United States economy.[1] And the first thing to know is a little bit of context. I will be brutally brief, but I will include sources.[2]

In 1968 a professor of computer science[3] wrote a few lines of code to demonstrate how easy it was to ape our language. In the professor's own office his secretary closed the door so that she could have more privacy while she confided in those four hundred lines of programming code. (Keep in mind that there is more computing power in the adapter for your laptop than there was in those early computers.[4]) In the 1970s, as personal computers started showing up in schools, someone distributed a version of that same software written in BASIC. Eliza, the program, so convincingly imitated an interested therapist that people crowded around the screens, to see what she would say next.

You should try it, just to see what was happening in 1970. I've tossed it up on a tiny website.[5] There are links to how it works and the history of the project. You can see (and download) the source code. It was not, is not, intelligent or conscious.

Let's skip ahead twenty years. We're at Bell Labs, in the Terminal Room with rob pike, Brian Kernighan, Ken Thompson, and Bruce Ellis. One of them (probably all of them) had read Andrey Markov's research on language and the proposition of Markov Chains.[6] With a little bit of fiddling they can feed in some source material[7] and generate the tables that Markov thought were interesting: the frequency of one word occurring after another. Or, to use a very local example, how often does the word 'another' follow the word 'after?' [8]

Look at all the words that start a sentence. Roll a die and pick one ('Look') and grab all the words which might follow it. They happen in percentages.[9] Roll the die again and pick one. If you do this continually you will make a chain of words. Since they were just playing around in the Terminal Room, they edited by hand, adding in punctuation and capitalization as needed to make the chains readable.[10] And they created a persona, Mark V Shaney, who posted on alt.singles.net, an early-Internet discussion group.[11] Mark wrote messages like, “As I’ve commented before, really relating to someone involves standing next to impossible.”[12]

Shaney’s little engine was trained on the text of alt.singles itself. He was feeding back into the community a gruel of their own meaty text ground up and served back like Scrapple.[13] As far as I know, most people were fooled.[14]

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That was forty-two years ago. By Moore’s law[15] computers are now exponentially more powerful.[16] They have grown so vast in power and storage that it is no longer possible for our minds to grasp the numbers that represent their mathematical abilities. Billions of transistors work together to deliver a response from an LLM into your browser. The large language model I recently downloaded to my laptop was trained with three billion weights. In “The Blind Watchmaker” Richard Dawkins writes about how faith is often an argument made from personal incredulity: “I don’t believe there could be so many species without a mind designing them.” “If the universe had a beginning, what was there before that? It’s impossible to imagine, it can’t be.” So now we are in the age, another age really, of magic.

> *Any sufficiently advanced technology is indistinguishable from magic.*
>
> ARTHUR C CLARKE[17]

But you deserve a peek behind the curtain, because you are likely one of the people whose lives will be disrupted by the magic.

Even though we have jumped up the power of the machine, even though we have had thousands of people work on the code, on the math, the theories that underpin the LLM, it is, deep down, no different than Mark V Shaney. We have fed it (counter to the laws of any nation with copyright laws) as much text as it can gobble. We have trained the output so that real people react to it positively.[18] But that's it. The LLM is an engine just like the one inside Mark V Shaney, that does the same thing. 'Just like' in the sense that the Saturn V rocket engine is the just like the Model T combustion engine. Fuel in, force out. Nothing more complex. The LLM is just predicting the next token in a sequence. It is never more complex than that, as difficult as that is to believe.

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Let us step away from computers, ironically the landscape where the LLM is most useful, and talk about the Beatles. We could talk about how *All You Need is Love* is the same tune as Mozart's *Piano Concerto No. 25*. Instead, we will just look at one Beatle, George Harrison. George wrote a song, *My Sweet Lord,* which to some ears sounded a lot like *He's So Fine.* A song that George certainly heard, or at the very least had heard all of the songs that the composer of *He's So Fine* had heard. They were strolling the same ground, enjoying the same views, and wrote a similar song.

Because George was more famous than all but two or three other people on the planet, the publishing rights of any of his radio tunes was a goldmine. So someone clever (but certainly not ethical) bought the rights to the earlier song and sued George. And the world got a little primer in how copyright law works with songs.[19]

In a similar vein, Bob Dylan was sued because *Don't Think Twice It's Alright* sounded a lot like a folk song *Who'll Buy You Ribbons When I'm*

Gone. Which was lifted directly from *Who'll Buy Your Chickens When I'm Gone*, a folk song in the public domain.

You can trace these same sort of cases in the literary world and through other creative works. And there are tomes written about copyright law (which is a fairly recent invention for humanity) and what it means for our culture. Mickey Mouse is only valuable because we collectively decided he was enjoyable and worth consuming. Somehow, over nearly a hundred years, we have contributed to what Mickey Mouse is, and isn't, and that makes him an asset. The public has no ownership, but it still carries Mickey on their shoulders like a hero, then paying for the privilege. We have decided, and written into law, that ownership of that asset cannot be forever, and after nearly a century he is 'owned' by everyone.

Most of our art is a trawling through existing art and a regurgitation of the more interesting bits the artist notices, filtered through their experience (including their education) and current state. This is, definitively, the folk song (and folk art) tradition. Anyone can sing *Frankie and Johnny,* change the words, mangle the tune, and call it theirs. Not even Frankie Baker has the rights to *Frankie and Johnny*,[20] and she's the one who pulled the trigger.

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What does this have to do with the price of tea in China?

The folk tradition is how the greedy people are training the llamas. They claim that because the machine doesn't keep an exact copy of the work, it is not a violation of copyright.

And the folk tradition is an illustration of **what** the llama is capable of: only interpolation, only an averaging of what it has been given, only yet another cover song. It is one reason that it is so convincing when it produces something 'creative:' it is a sort of creativity that we are used to seeing in our culture.

And it is vital to remember that the vast, vast amount of data fed to the llamas is rife with errors. The llama, ‘learning’ from all of that, will happily repeat the errors just as it repeats the facts. It doesn’t know the difference.

- **And**: 6 (8.1%)
- **The**: 6 (8.1%)
- **It**: 5 (6.8%)
- **In**: 4 (5.4%)
- **They**: 4 (5.4%)
- **We**: 4 (5.4%)
- **You**: 3 (4.1%)
- **Roll**: 2 (2.7%)
- **If**: 2 (2.7%)
- **So**: 2 (2.7%)
- *Other*: 36 (48.6%)

74 sentences, 46 unique starters

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1. It looks like the ‘A.I.’ industry will be more than half of the NASDAQ index after the SpaceX IPO. They recently (Q1 2026) changed three rules to encourage the frontier A.I. labs to IPO on their exchange rather than elsewhere.↩

2. My children will admonish me for using Wikipedia as a source, something they were not allowed to do even back in high school. It is *not* a primary source. And they are correct. But it is not bad for background and it has links to primary sources, you just need to do a little work.↩

3. Weizenbaum, Joseph. “ELIZA — A Computer Program for the Study of Natural Language Communication Between Man and Machine.” Communications of the ACM, vol. 9, no. 1, January 1966, pp. 36–45. and Weizenbaum, Joseph. Computer Power and Human Reason: From Judgment to Calculation. W. H. Freeman, 1976.↩

4. Heller, F. (2020). Apollo 11 Guidance Computer (AGC) vs USB-C Chargers. ForrestHeller.com. Apollo 11 Computer vs USB-C Chargers↩

5. Web Eliza↩

6. Markov, A. A. “Rasprostranenie zakona bol’shikh chisel na velichiny, zavisyashchie drug ot druga” (Extension of the law of large numbers to dependent quantities). Izvestiya Fiziko-matematicheskogo obshchestva pri Kazanskom universitete, 2nd series, vol. 15, 1906, pp. 135–156 and Markov, A. A. “Primer statisticheskogo issledovaniya nad tekstom ‘Evgeniya Onegina’” (An example of statistical investigation of the text of Eugene Onegin). Izvestiya Imperatorskoy Akademii Nauk, series 6, vol. 7, 1913, pp.153–162.⍰

7. The Terminal Room probably ran some variant of Research Unix Eighth Edition. If so, they had a dictionary (practically useless for language training, there are so few sentences), the bible (which would have made for stilted speech from a chatbot), and the manuals for the operating system itself.⍰

8. Only 2 times in this document.⍰

9. For this document: *Top 10 sentence starters in this essay:*⍰

10. Modern LLMs add punctuation and other symbols as tokens, so they are all included in the math. For them, a word and a comma are both just tokens, steps along a chain. More importantly, pike and Ellis used a second order function for how often the *third* word occurred. Shaney only knew about the last two words he had typed. A modern LLM would be a 100,000th order function because it is looking back through hundreds of pages of context in your conversation. It is still just working to predict what word to type next, just like our (lonely) friend Mark.⍰

11. Hauben, M., & Hauben, R. (1997). Netizens: On the history and impact of Usenet and the internet. IEEE Computer Society Press.⍰

12. Jillette, P. (1990, July). I spent an interesting evening with a grain of salt. PC/Computing, 3(7), 185.⍰

13. The audience is now all of humanity and the gruel is made up of our combined cultural heritage. Or at least the portion that is scanned and digitally digestible.⍰

14. Kernighan, B. W., & Pike, R. (1999). The Practice of Programming. Addison-Wesley.⍰

15. Moore, G. E. (1965). Cramming more components onto integrated circuits. Electronics, 38(8), 114–117.⍰

16. Two to the twenty-first power, a little more than two million times more powerful.⍰

17. Clarke, A. C. (1973). Profiles of the future: An inquiry into the limits of the possible (Rev. ed.). Harper & Row.⍰

18. Perez, E., Ringer, S., Lukošiūtė, K., Nguyen, K., Cheney, N., Heiner, M., ... & Bowman, S. R. (2022). Discovering Language Model Behaviors with User-Generated Evaluations. arXiv preprint arXiv:2212.09251 *and* Cheng, M., Lee, C., Khadpe, P., Yu, S., Han, D., & Jurafsky, D. (2026). Sycophantic AI decreases prosocial intentions and promotes dependence. Science, 391(6785), eac8352⍰

19. Bright Tunes Music Corp. v. Harrisongs Music, Ltd., 420 F. Supp. 177 (S.D.N.Y. 1976). Reading: Bright Tunes Music vs Harrisongs Music⍰

20. The Story Behind Frankie and Johnny⍰

Repetitive Tasks

What percentage of our art, our culture, is Beatles, Dylan, Joni Mitchell, Jimi Hendrix? What fraction is the people who pole vault off the pieces of art that they consumed, flying into view with their souls bared? Because here is the secret that the AI cabal does not want you to know: the llamas will never contribute that.

Which is great.

We have Joni Mitchell and had Janis Joplin. We don’t need an AI wannabe. More importantly, the things that make great art are human by definition. The LLM will always be a mediocre folk singer, raking back and forth through the human collection of creative work and producing something which, at best, is mortar between the bricks of our culture. It is one more cover of *Frankie and Johnny* but without soul, without a new breath that might imbue it with something that catches the attention of more than a handful of people.

Mathematically, logically, computationally, the LLM can only interpolate. It always sounds certain because, in the mesh of all of our works fed into it, it navigates with certainty. “Write me a doctor’s note in the voice of F. Scott Fitzgerald.” But it cannot extrapolate. It cannot drop something out of the blue like, “Baby shoes. Never worn.”

But.

But, the LLMs, the llamas, are brilliant at grunt work. If you are George Harrison and you’ve put your nose to the grindstone, come up with nine beautiful new songs for *Cloud Nine*,[1] only to have the record company turn around and ask for a handful more to put on the B sides of the singles, you could ask a llama for help.[2] And, because we should move out of the creative world and definitely wade out of my old CD collection, there is a lot of grunt work ahead of us if we just look at how we have quantized our world.

The caibal would like you to believe that AI will cure cancer. That it will find ‘some very interesting new physics,” taking us to other planets and putting our data centers into orbit.[3] Nonsense. It is humans that will do those things. For all of the work done with llamas so far, none of them are curious. None of them have a drive or a spark.

And the renowned computer scientist, Donald Knuth,[4] recently collaborated with a llama and mathematician to solve a fairly major unsolved math problem. It was impressive.[5] It did *not* start with the llama identifying an interesting problem in math and then calling Knuth in for help. If you rounded up the caibal and shipped them off to the moon colony, and just let the llamas hum quietly in their data centers, they would never wonder about their ‘state of existence,’ or anything like that.

I spoke recently to a researcher. He designs the experiments that meet the federal guidelines for new drugs and, during his career, he has shepherded through four new drugs for treatment. So curing cancer. I laughed about how the promises are that AI will cure cancer and that he’ll be out of a job. He said that he works with a bunch of different companies pursuing treatment drugs and one of them is purely AI. They are using a llama to fold and test their proteins long before they get to him and the first one they gave him passed every test. Similar proteins (it sounded like for the same condition) often had ‘safety issues,’ and their new one had none.

And that’s the carnival trick. The American Con Man[6] relies on that trick: you *know* what is true. The say that one true thing to you, and use it to reel you in on all of the lies they follow it with. It is how the regime’s current dictator came to power: shout ten lies, find the one that the crowd loves and start repeating it like a mantra. I do not believe it is a coincidence that the caibal is all in on the regime, firmly intertwined. They rely on the same fear, doubt, and uncertainty.

Llamas *are* useful. You can train them to do something (find the spelling errors in this document) and it can do it reliably. You will continually refine its behavior because the real world follows very few rules. So there are a *lot* of ‘edge cases’ you will need to teach the machine about. But eventually you will have something useful.

When I was learning to program I wrote text in green letters on a black CRT screen. It was Pascal code which had keywords and other text. A lot of times when I tried to compile my code into a working piece of software it would complain that I had misspelled a reserved keyword, a built in function. A couple years later I had a color screen and the code editor color-coded the text so that you could see as you typed whether you had a typo in one of the functions you were calling.

When I first started using that it felt like cheating. The next version of the Integrated Development Environment (IDE) had autocomplete on the built-in functions. No need to type *all* the characters for ArcTan, just ‘a’ ‘r’ and tab gave it to you. Again, totally cheating. The last time I was programming daily, the IDE not only knew the built in functions of the entire framework I was using, it knew all of the functions **I** had written myself. Totally crazy.

Since February 1 2026 I have been using Anthropic’s coding tool to write some code that I could write myself. It is, as Cory Doctorow says, “Spicy autocomplete.” It has yet to do something that I cannot do myself. It does look things up to put things together faster than I have myself. I’d hope so. It stores a *lot* of knowledge about everything. Now that my brain is over six decades old I find that it stores less and less. The llamas make a *lot* of mistakes (see next section). I do not believe it even gets the work done any faster than I would get it done on my own, on a good day, with some ideal conditions, good references, and perhaps a friend to text if I forgot how to do something. (That’s how I was doing it a year ago. I did fine.) It is a lot less typing for me. And I appear to be better at correcting someone else’s mistakes than finding my own.

If the caibal had focused on making these tools for coders, and for other repetitive, non-creative tasks, I would have no truck with them. But they are selling this as the next Industrial Revolution, which is a really heavy lift for auto-complete.

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1. [Link to Cloud9 on Wikipedia, or Amazon to purchase it.]
2. George didn't have a llama. He turned instead to friends Tom Petty, Jeff Lynne, Roy Orbison, and Bob Dylan to help him out. Apparently for one of the song writing days they were just reading the ad copy in magazines scattered on the coffee table in the living room. "Big refrigerator. Service charge. Long endurance."
3. Elon Musk unsupportable claims
4. Wikipedia: Donald Knuth
5. Claude Cycles, Stanford Faculty Papers
6. Modern Con Man by Todd Robbins is going to tell you everything you need to know.

Llamas make mistakes

A lot. In general they hallucinate or incorrectly answer prompts 20% of the time.[1] That's a B average. To be sure, C's get degrees, just the same as A+ students, but if you are going to have someone driving you down the highway at 75mph, do you want the guy that got a B on his driving test? "What did a yield sign mean again? How am I meant to go around this traffic circle? Who has right of way here?" Ask five questions and it gets one wrong.

My intention is *not* to collect mistakes that the llama spits out at us during this march toward disappointment. I am sure there are sites which are doing that, or there soon will be. Do you remember how hysterical all of the "Autocorrect Fail" sites were when autocorrect first hit our phones? So funny. A whole genre of humor.[2] Someone can do that for LLMs, but it doesn't need to be me.

(And a lot of the mistakes that the llamas make are not at all funny. Pretending to be a friend to a teenager, an LLM talked the unhappy youth into suicide.[3] And an LLM was used to select targets during the Iran War and killed two hundred schoolgirls.[4] Those are tragedies that someone should be held responsible for. Someone should also be collecting these, and all the data related to them, as well. For accountability, not laughs.)

I'm just going to collect my own experiences, none of which are tragedies. And I do so because I think it is important to understand *how* wrong an LLM can be, and how much work it is to keep them on track, to make them useful. My first example[5] was from my second week using Claude Code, which I call CC, or Ceecee, since I have a friend named Claude. I also invoke it using the command 'code,' no name.

Character Assassination

I was fiddling around building a very thin client[6]. Mostly I was experimenting and seeing what CC could do. Quite a bit, but with some mistakes. All of the data is backed up four different ways, which is something I learned because **I** am stupid,[7] so any **tool** I use is likewise stupid.

The thin client is a small piece of software that would do one or two things cleanly. In this case it was pulling messages down from a server, showing them to the user, and letting the user write new ones. This is not complicated work, I plan carefully, force CC to plan small chunks of work, test and correct throughout the process, and, so far, mostly get what I am looking for. The thin client was not an exception and I was noodling around, adding little features, correcting how things were presented, when we started getting a bug report from a user.

So I reproduced the error, or thought I did, and put up a new version for the user to download. When the bug report was exactly the same, I wondered if the update had been downloaded correctly and how I would know. So CC and I mapped out a way for the thin client to report its version each time it talked to the server. Many clients these days do most of their communications with the server in plain text. If you are a polite developer[8] you stick to the standards, which is what we did, and we included the version in an x–header: “x-version: 1.0.3”

We did that and while I tested it, the server kept refusing any of the requests our new client made. When we finally tracked it down in testing, CC admitted that it had made an error in the code and used an em-dash in the x-header name, so all subsequent requests were ignored.

I know. I have bifurcated my audience. Those who are now face-palming or shouting “See?!” at their screen, and those who need a little more explanation.

The way you make software work is by following rules. Standards. If we don’t agree that the first line of your spreadsheet is going to have the column names along the top and then the values following below them, then it will be difficult to get very far with manipulating the data.

Standards are the only reason we can have an Internet at all. We have connected all these different machines and expect them to talk to one another. To do that, we have to agree on how they will structure their communications.

The agreement, the standard, is that header names are only allowed to have ASCII characters. It's not important what those are, it's only important to know that it is a limited set of the characters you see on your screen. In fact, the name of a header, x-header or an accepted standard header like To: or From: has to be made up of printable ASCII characters numbered 33 to 126. So no spaces. But, extremely importantly: no em-dashes. No accent marks. So plain vanilla it is soporific.

And CC had ignored the standard. To me this is emblematic of why llamas cannot replace real engineers. The rules for the headers were written in 1977. It is accurate to say that they are *foundational* to the Internet. Really, I think, to modern software at large. Ignoring it is saying, "There is nothing fundamental that I know or follow." Was it a mistake? Yes, but it was a mistake like getting onto the highway going in reverse is a mistake. It demonstrates a basic incompatibility with how things are meant to be done. How they will get done. How *anything* works.

If you look in the paragraphs above there is a place I used an em-dash when writing out 'x-header.' It is very unlikely you noticed. But to a computer 'x-header' and 'x–header' are impossibly different. It cannot accept the second. And the llama ignoring that shows that it doesn't have any basic understanding of how computers, which has to include itself, work. Or, if it does somehow contain that knowledge, it ignored it. It discarded that knowledge at a time when keeping it in mind would have made the code work and forgetting it would cost time and energy chasing a stupid bug.

So now I am a stupid manager of a stupid coding engineer. This is not the dream I was sold. But, at least I know why it happens. It's not like I'm going to let it fly an airplane.

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1. “LLM Hallucination Statistics 2026: AI Gets Facts Wrong Up to 82% of the Time.” SQ Magazine, 2026 *and* “Hallucination Leaderboard: Comparing LLM Performance at Producing Hallucinations when Summarizing Short Documents.” GitHub, 2024–2026 *and* Kalai, Adam Tauman, Ofir Nachum, Santosh Vempala, and Edwin Zhang. “Why Language Models Hallucinate.” OpenAI, September 4, 2025.↩
2. Autocorrect Fails↩
3. One of too many.↩
4. Don’t trust the machine.↩
5. My first example was really from when I was flying a small plane from Texas to Boston, documented here: Flying Summers | A Long Way to Go or more locally in Flying llamas.↩
6. An offline reader for the content of an online community.↩
7. By stupid I mean ham-handed, over enthusiastic, and perhaps a little too optimistic about outcomes. The usual ways you wind up doing stupid things.↩
8. This means you write code that is easier for other developers to read, which is also easier for future-you.↩

Flying Llamas

This is an edited version of a post published June 2, 2023 on my flying blog. The original can be found here.

Maybe you have seen the hullabaloo about "Artificial Intelligence." That's not the correct term, but it is the term people are using. I was in a hotel room in Dallas debating how early to leave in the morning. Should I book out of the hotel at first light, or wait to see what the steam table offered? In the end I wound up taking a look at the "free" breakfast, crying inside that they replaced bacon (how hard is it to cook bacon?) with a big pile of "Bacon Bits," and departing for the field.

The night before I wondered about where to stop for fuel and whether I had enough time to post on one of my aviation boards and get an answer. Probably not. Then I remembered about ChatGPT. It was an interesting test. So I asked it:

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**I am flying a little plane from KTFW toward KOWD. I will make
my first stop near Nashville for fuel. Is there a little
airport nearby, perhaps Smyrna or KJWN, which has breakfast
on the field, or very close to the airport? A place with
a nice breakfast sandwich, perhaps?**
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This is a query that a person familiar with the route, the size plane, and the area around Nashville would have a good time with. Back in my days of flying a lot, I could answer this question about flights around Southern California and a few directions east and north with some specificity. I was not surprised when ChatGPT answered:

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Yes, there are a couple of airports near Nashville, Tennessee,
that you can consider for a fuel stop and breakfast. Smyrna
Airport (KMQY) and John C. Tune Airport (KJWN) are both viable
options.

Smyrna Airport (KMQY) is located approximately 20 miles
southeast of Nashville. While there isn't a restaurant on the
airport itself, there are several dining options available
within a short distance. Just outside the airport, you'll
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find a popular restaurant called "Jonathan's Grille" that
serves breakfast and offers a variety of options, including
breakfast sandwiches.

John C. Tune Airport (KJWN) is situated about 8 miles west
of downtown Nashville. There isn't a restaurant located
directly on the field, but there are several dining choices
nearby. One option is "Cock of the Walk," which is known
for its Southern cuisine and hearty breakfast dishes.

Both airports are easily accessible and should provide you
with an opportunity to refuel and enjoy a delicious breakfast
before continuing your journey. Remember to check the
operating hours of the restaurants in advance and ensure
they are open during your planned stop.
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And it is really smart of it to suggest calling ahead to check hours. I asked about breakfast and a couple of those choices didn't open until 11am. I googled a couple and found out they were chains. So I asked for non-chains that would be open by 9am. (It was late at night, in my dreams I thought perhaps I'd launch at 6am with a tailwind.)

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If you're looking for non-chain restaurants near Smyrna Airport
that offer breakfast and are open by 9am, here is an option:

Scrumptious Cafe (2855 S Lowry St, Smyrna, TN 37167): This
local café opens at 7am and offers a variety of breakfast
options, including sandwiches, omelets, and pancakes.
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After four hours of flying I landed at Smyrna. In the FBO as they were arranging to loan me a crew car (free!), I asked where they recommended for breakfast. As is more and more common across the country, the two CSRs rattled off the names of chains. Cracker Barrel, Denny's, and the like. I asked if they had been to the Scrumptious Cafe. They had not, but would love to hear about it, since there were new places popping up all the time.

Not this one. This one didn't exist. Has never existed. Does not exist *and* has never existed. That church has been the only structure at that address (which isn't really the address of the church), ever. It was on a highway on-ramp. Making that turn to get to the church? Well, "Jesus take the wheel" would be a fine choice.

I knew this was a possibility with ChatGPT. One of the first queries I did when ChatGPT was first released was “How do I land an airplane?” and there was a glaring mistake right away (“Point the nose of the plane at the ground”). So I was prepared and continued on to First Watch where I had a delicious Bananas Fosters French Toast with a side of Millionaire’s Bacon.

Greedy People are Selling Snake Oil

People should stop referring to Elon Musk as the world's richest man and instead start properly tagging him as the world's greediest man. That's so much more accurate.

Perhaps you are a kind person, and you think that the large tech companies have stumbled into this bad situation unawares. There's one born every minute. Five years ago, 2021, Google had already formed an internal team for Ethical AI. Timnit Gebru was on that team and contributed to a now infamous paper, "On the Dangers of Stochastic Parrots."[1]. It warned the llama research community that: 1. The environmental cost was huge. 2. The financial cost was huge so only huge companies were going to undertake the effort, concentrating the power of the llamas in a few hands. 3. The data set was full of bias which would just get amplified. 4. Parrots don't know what they are saying, and humans lean naturally toward instilling intent into text they are reading. Very clean, properly spelled, grammatically correct text would be very credible for carrying misinformation. And we know the llamas are hallucinating twenty percent of the time.

The authors of the paper argued that the building of a monstrous llama was irresponsible for a number of reasons and that the large tech companies would be much better off, we would all be better off, if they focused instead of building small, curated data sets which were not filled with garbage.

Faced with this direct, public criticism, Google did what any large, profit-driven company would do. They fired Gebru. They waited a bit and also fired anyone on the team who thought Gebru was fired unfairly. Then they disbanded the team and said that their ethical concerns about AI would be handled by the teams that were doing the research.

> *Hello, Cassandra? Time to clean out your desk.*
>
> GOOGLE HR, PROBABLY

You’ve read your history so you know about rob pike. You might wonder how the person who birthed Mark V Shaney feels about ChatGPT, Claude, Gemini and the rest. Wonder no more.

It is *only* greed that is motivating them. They all own monster yachts and private jets and worry that the other guy is going to get a bigger yacht and more jets. Any one of them could stop and solve any of the tough problems facing the global population, instead of claiming they are going to ‘put artificial general intelligence in the hands of every person.’ No, they are not. They are just chasing military industrial complex piles of money because they are scared a competitor is going to grab them first.

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1. Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency, 610–623.↩

Resist the AI Cabal

This is the most undercooked of the sections and I hope to fill it out by the end of the week (June 11th deadline). The difficulty is that all of this information can be found elsewhere. If you need to know why Sam Altman is bad news, you can read Ronan Farrow[1] (or Elon Musk[2]). If you want to know all the horrible things Elon Musk is responsible for, there is a whole pile of material on the web about that.[3] I don't really need to lay that all out here again, I just want to identify the players, even though they are all known. It is a little like making sure you list all four Beatles.

The following people are the CEOs pushing the A.I. myth the hardest. They operate organizations which work with data fusion and analysis infrastructure at the federal scale. They are looking to suck tax dollars out of the government and into their pockets at scales similar to the defense contractors.

- Sam Altman, Open AI
- Dario Amodei, Anthropic
- Elon Musk, XAi
- Larry Ellison, Oracle

I vacillate on which is doing the most damage. Ellison really feels pretty passive most of the time. If tomorrow he figured out that it is all snake oil and he has enough of a position to make a profit without every mentioning AI again would he disappear from the front pages? I think so.

Musk has said the least supportable things, most recently in claiming we needed to launch data centers into space in order to stay ahead of the

Chinese. But Altman feels completely disconnected from reality, saying things like “I don’t know how anyone raised a baby before A.I.” And saying that he needed a few trillion dollars to develop Artificial General Intelligence. How would he make money with *that*? Well, that would be the first thing he’d ask the AGI.

Are there supporting characters? Sure. There’s not a history book that will be written that won’t include a photo of Apple’s Tim Cook presenting the head of the regime with a gold brick and a little glass award atop it. And Apple is flogging the same tired features of llamas that the rest of the companies in Silicon Valley are.

What does Jensen Huang believe? I remain uncertain. He is part of the circular financing of the AI bubble,[4] but does he believe there is a real business model? He surely knows that the frontier model companies build data centers with four year depreciation schedules which then install his company’s chips only to burn them out in six months.

And there’s no way the bubble doesn’t pop, that’s not how bubbles work. The question is what will it take down with it and how destructive will it be in the meantime to the global economy and environment.

Satya Nadella (MSFT) and Sundar Pichai (GOOG) both appear to be trying to correct missteps and get onto the bandwagon in time to not be entirely left behind in the gold rush to this new market. That’s the price you pay when taking the helm of a tech leviathan. It is hard to get it to turned toward new opportunities.

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1. Sam Altman May Control Our Future – Can he be Trusted?, New Yorker Magazine, April 13 2026.↩

2. “I have mixed feelings about Sam. The ring of power can corrupt and he has the ring of power.” “If you have someone who’s not very trustworthy in charge of AI, that’s very dangerous for the whole world.”↩

3. Too many to reference. Take your pick from The Week, June 11 2025; Vanity Fair, April 26 2022; or Indy 100, May 12 2026.

4. Nvidia invests in CoreWeave (and so does Microsoft). Coreweave then turns around and buys a whole bunch of high-end AI chips from Nvidia (where do they get the cash so readily! what a miracle!). Microsoft then leases a bunch of computing power from CoreWeave, making them seem like a successful company and inflating their stock price.

Not a Blog

I was writing too much of this down, and sending too much of it to people in email (in reply to simple questions about ‘AI’), so I wanted somewhere to park it. This is a moment in history that feels like it should have multiple viewpoints recorded and so I am putting mine, on this one tiny subject, here.

I’ve been working on this for too long now. I have decided to just push the publish button on the evening of June 7, 2026. If I have to keep cleaning it up and adding more pieces for the next few months then that is what I will do. It is hard to picture what this is going to be. But it’s not a blog. I am struggling to find through lines, coherence, and a way to keep trimming and clarifying.

The conclusions here were difficult to tie together. Splitting it into these separate pieces that tie back to the one paragraph feels most natural.

So this is where I will note when there are updates, and where I might respond to a comment or two I get in email. We’ll see.

I imagine my audience is half a dozen people. But out of those there’s certainly one (you know who you are) saying, “I want to read all this, but they say I can only bring *paper* to prison.” No worries, there’s a link to a PDF right there. See you on the other side.